Università degli Studi di Cagliari

Master course in Computer Engineering, Cybersecurity and Artificial Intelligence

Network Security Project

Statistical and Machine Learning-Based Detection of DoS/DDoS Attacks in IoT Network Traffic

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Author: Andrea Labelli

Professor: Marco Martalò

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Abstract

Denial-of-Service (DoS) and Distributed Denial-of-Service (DDoS) attacks remain among the most critical threats to modern networks, particularly in Internet of Things (IoT) environments, where resource-constrained devices are highly vulnerable to flooding-based attacks. This work presents a detection framework that combines statistical traffic analysis and supervised machine learning techniques to identify DoS/DDoS activity in network traffic.

The proposed methodology first performs a detailed statistical characterization of TCP and UDP flows on the TON_IoT_Train_Test_Network dataset, analyzing behavioral and volumetric features such as connection states, packet counts, and byte distributions in order to highlight discriminative patterns associated with flooding behavior.

In addition to the binary comparison between DoS/DDoS and other traffic, an attack-type comparative analysis is conducted across multiple categories (Scanning, Brute Force, WebAttack, and Ransomware) to better understand how flooding-based attacks differ from other intrusion scenarios. This comparative study confirms that DoS/DDoS attacks exhibit predominantly volumetric patterns, while other attack types generally involve more targeted and lightweight communication behaviors.

Three supervised learning models — Random Forest, Logistic Regression, and Support Vector Machine — are then evaluated for binary classification between DoS/DDoS and other traffic.

To address the class imbalance present in the dataset, the Synthetic Minority Oversampling Technique (SMOTE) is applied to the training data in order to generate synthetic samples for the minority class and improve the learning process of the classifiers.

The experimental results demonstrate that ensemble-based models, particularly Random Forest, achieve superior performance in terms of detection capability and robustness to class imbalance. However, the results also show that detection performance is influenced by dataset imbalance and feature complexity, and that no model achieves perfect classification.

Overall, the combination of interpretable statistical analysis, comparative attack profiling, data-driven machine learning, and data balancing techniques provides an effective yet realistic approach for DoS/DDoS detection in modern network and IoT security scenarios.

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1 Introduction

In recent years, networked systems have become a fundamental component of modern society, supporting a wide range of digital services such as cloud computing, online banking, e-commerce platforms, healthcare systems, and industrial control infrastructures.

The rapid expansion of the Internet and the growing adoption of Internet of Things (IoT) devices have significantly increased the scale and complexity of modern network environments. While the interconnection of billions of devices enables unprecedented levels of automation, scalability, and accessibility, it also introduces significant security challenges. IoT devices are often characterized by limited computational resources, heterogeneous architectures, and sometimes insufficient security mechanisms. [1]

As a consequence, these devices may become attractive targets for malicious actors or be exploited as part of larger cyberattack infrastructures. One of the fundamental objectives of information security is to ensure the availability of services, alongside confidentiality and integrity. However, modern network infrastructures are frequently exposed to attacks aimed at disrupting service availability by overwhelming network or system resources.

Among these threats, Denial of Service (DoS) and Distributed Denial of Service (DDoS) attacks represent a major concern due to their ability to degrade or completely interrupt the operation of network services. The growing scale of IoT deployments further amplifies this risk, as large numbers of poorly secured devices can be compromised and organized into botnets capable of generating massive volumes of malicious traffic.

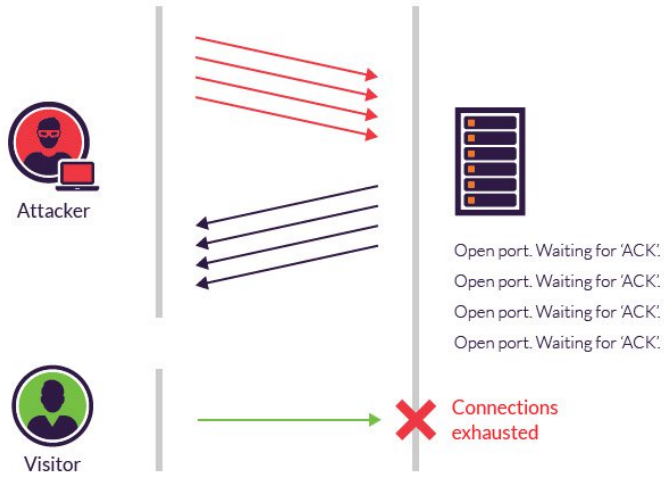

Figure 1: SYN Flood attack example

As a result, detecting abnormal traffic patterns and identifying potential attacks has become a critical task in modern network security. Traditional

intrusion detection approaches often rely on rule-based systems and statistical thresholds derived from traffic characteristics such as packet rates, connection counts, and protocol distributions.

While these methods are generally efficient and interpretable, they may struggle to adapt to the dynamic and heterogeneous nature of modern network traffic, especially in IoT environments.

To address these limitations, machine learning techniques [3] have increasingly been explored for network intrusion detection. By learning patterns directly from data, machine learning models can capture complex relationships between traffic features and malicious behaviors, potentially improving the detection of both known and emerging attacks.

Nevertheless, these approaches also introduce new challenges, including the need for high-quality labeled datasets, issues related to class imbalance, and concerns regarding model interpretability. In this context, combining statistical analysis with machine learning-based methods represents a promising direction for improving the detection of anomalous behaviors in network traffic, particularly within IoT environments.

1.1 Denial of Service (DoS) Attacks

A Denial of Service (DoS) attack aims to deliberately disrupt the normal operation of a networked system or service by exhausting its computational, memory, or communication resources. By overwhelming the target with a large volume of requests or malformed packets, an attacker prevents legitimate users from accessing the service, thus violating the availability property of information security. Unlike attacks that seek to steal or manipulate data, DoS attacks focus primarily on making services unreachable.

In a traditional DoS scenario, the malicious traffic originates from a single source or a limited number of machines. Although this characteristic may simplify detection compared to distributed attacks, a well-crafted DoS attack can still cause significant degradation of performance, especially when it exploits protocol weaknesses or resource-intensive operations. Common examples include ICMP flooding, UDP flooding, and TCP SYN flooding, where attackers generate massive numbers of packets or half-open connections in order to consume server resources.

DoS attacks are particularly effective against services that are not adequately provisioned or protected by rate-limiting and traffic filtering mechanisms. As modern networks increasingly rely on real-time communication and cloud-based infrastructures, even short-lived DoS attacks can result in noticeable service disruption and economic impact.

1.2 Distributed Denial of Service (DDoS) Attacks

Distributed Denial of Service (DDoS) attacks represent an evolution of DoS attacks in which the malicious traffic is generated by a large number of compromised devices rather than a single attacker. These devices, often infected with malware and remotely controlled, form what is commonly referred to as a botnet. By coordinating thousands or millions of sources, attackers can generate traffic volumes that far exceed the capacity of most networks and servers.

The distributed nature of DDoS attacks significantly increases their effectiveness and resilience. Blocking a single IP address or small set of addresses is insufficient, since attack traffic is spread across many geographically dispersed sources. As a result, mitigation strategies must rely on large-scale traffic analysis, filtering, and often the support of specialized infrastructure such as content delivery networks (CDNs).

In addition to direct flooding attacks, some DDoS strategies rely on amplification and reflection mechanisms. In such scenarios, attackers exploit legitimate network services that respond with larger packets than the original request. By sending spoofed requests to these services using the victim's IP address as the

source, attackers can generate significantly amplified traffic directed toward the target system.

DDoS attacks are among the most frequently observed large-scale cyber threats and are often used as tools for extortion, political protest, or competitive sabotage. Their prevalence makes them a primary focus of modern intrusion detection and network monitoring systems.

1.3 Categories of DoS/DDoS Attacks

DoS and DDoS attacks can be broadly classified according to the layer of the network stack they target and the techniques used to overwhelm the victim's resources. Understanding these categories is important for designing appropriate detection and mitigation strategies, as different attack types exhibit distinct traffic characteristics.

- Volumetric attacks, which aim to saturate network bandwidth by generating extremely high traffic volumes. These attacks attempt to exhaust the available network capacity, preventing legitimate packets from reaching the target. Examples include UDP floods and ICMP floods, where large numbers of packets are transmitted at high rates toward the victim system.

- Protocol-based attacks, which exploit weaknesses in network or transport layer protocols. These attacks often target the mechanisms used to establish or maintain network connections, forcing the server to allocate resources for incomplete or invalid sessions. A well-known example is TCP SYN flooding, where attackers create a large number of half-open connections.

- Application-layer attacks, which target specific services such as web servers or APIs. Unlike volumetric attacks, these attacks may generate relatively low traffic volumes but involve requests that require significant processing by the server. HTTP floods are a common example, where attackers repeatedly request resource-intensive web pages or services while attempting to mimic legitimate user behavior.

Each category presents distinct detection challenges. Volumetric attacks may be identified through abnormal traffic rates or bandwidth usage, while application-layer attacks often require more sophisticated behavioral analysis to distinguish malicious activity from legitimate user interactions.

1.4 IoT Security Context and Vulnerabilities

The rapid proliferation of Internet of Things (IoT) devices has further amplified the relevance of DoS and DDoS attacks in modern network environments. IoT systems are often composed of a large number of heterogeneous devices with limited computational resources, minimal built-in security mechanisms, and long deployment lifecycles. These characteristics make IoT infrastructures particularly vulnerable to both direct attacks and large-scale abuse as traffic generators in distributed attacks.

Many IoT devices operate with constrained processing power, memory, and energy availability, which limits the adoption of advanced security mechanisms such as complex encryption schemes or continuous traffic inspection. In addition, insecure default configurations, outdated firmware, and lack of regular patching contribute to the exposure of IoT devices to compromise. As a result, compromised IoT nodes are frequently recruited into botnets and used as sources of malicious traffic in large-scale DDoS attacks.

The impact of DoS and DDoS attacks in IoT environments extends beyond traditional service disruption. In critical application domains such as smart cities, healthcare, industrial automation, and intelligent transportation systems, availability failures may lead to safety risks, economic losses, and severe operational consequences. Consequently, the detection of anomalous traffic patterns associated with DoS/DDoS attacks in IoT networks represents a key challenge for contemporary cybersecurity research.

Datasets such as TON_IoT have been specifically designed to capture these characteristics by modeling realistic IoT traffic and attack scenarios. By incorporating both benign IoT communication patterns and multiple cyberattack types, including DoS and DDoS, such dataset enables the development and evaluation of detection mechanisms that are representative of real-world IoT deployments.

2 Scenario

The experimental scenario considered in this project focuses on the detection of Denial of Service (DoS) and Distributed Denial of Service (DDoS) attacks within a heterogeneous network environment that contains both benign traffic and multiple categories of malicious traffic. The objective is to evaluate the effectiveness of different detection approaches—namely, statistical rule-based methods and supervised machine learning models—under realistic conditions in which legitimate and malicious traffic coexist.

The experiments are conducted using the "**TON_IoT_Train_Test_Network**" dataset [5], which is publicly available on Kaggle, an online data science platform that hosts datasets, machine learning competitions, and collaborative research resources.

Kaggle is widely used by researchers and practitioners as a benchmark source for developing and evaluating data-driven security solutions, providing access to curated datasets and reproducible experimental environments.

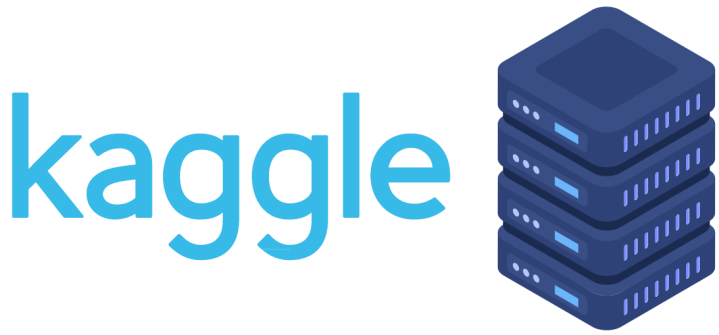

Figure 2: Kaggle dataset

The TON_IoT dataset was created to support research in intrusion detection and Internet of Things (IoT) security. It contains network traffic traces collected from a realistic testbed environment that simulates normal user behavior as well as multiple cyberattack scenarios. The dataset provides flow-level records enriched with statistical and protocol-based features, making it suitable for both traditional traffic analysis and machine learning-based detection.

Each record in the dataset represents a network flow characterized by attributes such as the transport protocol, connection state, number of packets exchanged between source and destination, and total bytes transferred. These features capture both temporal and volumetric properties of network communications, which are known to be strongly correlated with flooding-based attacks.

The dataset includes several labeled traffic categories, comprising normal (benign) traffic and different attack types, including DoS/DDoS, scanning activities, brute-force attacks, web-based attacks, and ransomware-related traffic. In this project, the original multi-class labeling is mapped into a binary classification problem. All DoS and DDoS samples are grouped into a single class representing availability attacks, while the remaining categories are grouped into an "Other" class. This transformation enables a focused analysis on the detection of flooding-oriented attacks.

Before performing the analysis, the dataset is randomly shuffled and divided into training and testing subsets in order to avoid bias and ensure reproducibility. In addition, a subset-based processing strategy is adopted, where flows are grouped into fixed-size blocks of 1000 samples each, that represent short observation windows of network activity.

Within this scenario, two complementary detection paradigms are investigated. First, a statistical characterization of traffic behavior is performed to identify patterns associated with DoS/DDoS activity, such as abnormal packet counts, byte distributions, and suspicious TCP connection states.

In addition, a comparative analysis across different attack types is conducted in order to highlight the differences in volumetric traffic characteristics between DoS/DDoS attacks and other categories such as scanning, brute force, web attacks, and ransomware.

Finally, supervised machine learning models are trained on the same feature set in order to automatically learn discriminative patterns from labeled data.

This experimental framework enables a systematic comparison between rule-based and data-driven detection approaches, providing insight into their relative advantages and limitations in modern network security environments.

3 Implementation

This section describes the methodology adopted to preprocess the dataset, select relevant features, and design machine learning models for the detection of DoS/DDoS attacks.

3.1 Data Preprocessing

The TON_IoT_Train_Test_Network dataset contains a large number of network flow records, each described by categorical and numerical attributes. In order to focus on traffic characteristics that are directly related to flooding-based attacks, a subset of features was selected for analysis:

- Transport protocol (*proto*)
- Connection state (*conn_state*)
- Number of source packets (*src_pkts*)
- Number of destination packets (*dst_pkts*)
- Number of source bytes (*src_bytes*)
- Number of destination bytes (*dst_bytes*)

Categorical attack labels originally provided by the dataset were mapped into two classes: **DoS/DDoS** and **Other**.

The latter includes normal traffic as well as other attack categories such as scanning, brute-force, web-based attacks, and ransomware. This transformation converts the problem into a binary classification task.

Before further processing, the dataset was randomly shuffled to remove ordering effects and ensure unbiased sampling.

The selected features capture both connection behavior (through the protocol and connection state) and traffic volume characteristics (through packet and byte counts).

3.2 Statistical traffic analysis

An exploratory statistical analysis was performed to study the behavior of DoS/DDoS traffic compared to other types of traffic.

For TCP traffic, the distribution of connection states was analyzed by mapping specific TCP states into two categories: Suspicious and Normal.

States associated with incomplete handshakes, rejected connections, or unexpected resets were classified as "Suspicious", as they are often observed during TCP flooding attacks such as SYN floods. Conversely, states corresponding

to successfully established or normally progressing connections were labeled as "Normal". This mapping simplifies the analysis by transforming the original multi-state representation of TCP connections into a binary categorization that highlights abnormal connection patterns.

Below is the connection_state mapping used:

TCP State	**Category**
S0	Suspicious
REJ	Suspicious
RSTO	Suspicious
RSTR	Suspicious
SH	Suspicious
SF	Normal
S1	Normal
S2	Normal
S3	Normal
OTH	Normal

Table 1: Mapping of TCP connection states

The TCP states reported describe the outcome or progress of a TCP connection attempt. States such as $S0$ indicate that a connection attempt was observed but no reply was received, which is often associated with incomplete handshakes. REJ represents connection attempts that were actively rejected by the destination host, while $RSTO$ and $RSTR$ correspond to connections that were reset during their establishment.

On the other hand, states such as SF, $S1$, $S2$, and $S3$ correspond to connections that were successfully established or progressed normally through the TCP handshake and data exchange phases. The OTH state represents other types of normal TCP activity that do not fall into the previous categories.

For UDP traffic, which does not provide connection state information, the analysis focused on volumetric features. The mean values of source and destination packets and bytes were computed for consecutive traffic subsets, separately for DoS/DDoS and Other classes.

3.3 Machine Learning-Based Classification

Three supervised machine learning algorithms were employed:

- Logistic Regression, which provides a simple yet effective linear baseline for binary classification problems. Logistic Regression models the probability that a given input belongs to a specific class by applying the logistic (sigmoid) function to a linear combination of the input features. During training, the model learns a set of weights that maximize the likelihood of correctly classifying the training samples. Despite its simplicity, Logistic Regression is widely used in classification tasks due to its interpretability and relatively low computational cost. Additionally, it allows the estimation of feature importance through the learned coefficients, which can provide insights into the relationship between network traffic features and attack behavior.

- Support Vector Machine (SVM), which aims to find an optimal decision boundary that maximizes the margin between different classes in the feature space. The SVM algorithm identifies a set of critical training samples, known as support vectors, that define the separating hyperplane between classes.

- Random Forest, which is an ensemble learning method based on the combination of multiple decision trees. Each tree in the forest is trained on a random subset of the training data and features, a process known as bootstrap aggregation or bagging [2]. The final prediction is obtained by aggregating the predictions of all trees, typically through majority voting in classification tasks. This ensemble approach reduces overfitting and improves generalization compared to a single decision tree. Random Forest models are also capable of capturing complex non-linear relationships among features and can provide estimates of feature importance based on how frequently features are used to split the data.

All models were trained using the same feature set and training data to ensure fair comparison.

3.4 Model Training and Testing

The dataset was split into training and testing subsets using an 80/20 ratio. Models were trained on the training set and evaluated on the testing set.

Predicted labels generated by each model were compared against ground truth labels in order to compute performance metrics.

3.5 Evaluation Metrics

The performance of the classification models was evaluated using three standard metrics commonly adopted in machine learning and intrusion detection research: Accuracy, Recall, and F1-score.

Accuracy measures the overall proportion of correctly classified samples among all samples in the dataset. It provides a general indication of the model's performance.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN}$$

where TP (True Positives) represents correctly identified DoS/DDoS samples, TN (True Negatives) represents correctly identified "Other" samples, FP (False Positives) represents "Other" samples incorrectly classified as DoS/DDoS attacks, and FN (False Negatives) represents DoS/DDoS samples incorrectly classified as "Other" traffic category.

Recall measures the ability of the model to correctly identify actual DoS/DDoS samples:

$$\text{Recall} = \frac{TP}{TP + FN}$$

The F1-score is the harmonic mean of precision and recall and provides a balanced measure between them:

$$\text{F1-Score} = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

4 Results

This section presents the results obtained from the statistical traffic analysis and from the machine learning-based classification models.

4.1 Statistical Analysis Results

The statistical analysis highlights clear and measurable differences between normal TCP traffic and connections classified as suspicious.

4.1.1 TCP Flooding Analysis

For TCP traffic, the percentage of connections labeled as "Suspicious" remains relatively stable across traffic subsets.

On average, approximately 30–35% of TCP connections fall into the Suspicious category, while 65–70% are classified as "Normal".

The computed TCP threshold is approximately 50%, which effectively separates abnormal connection behavior from normal traffic distribution.

This consistent separation indicates that TCP flooding attacks significantly alter the distribution of connection states, particularly increasing the presence of incomplete or reset connections.

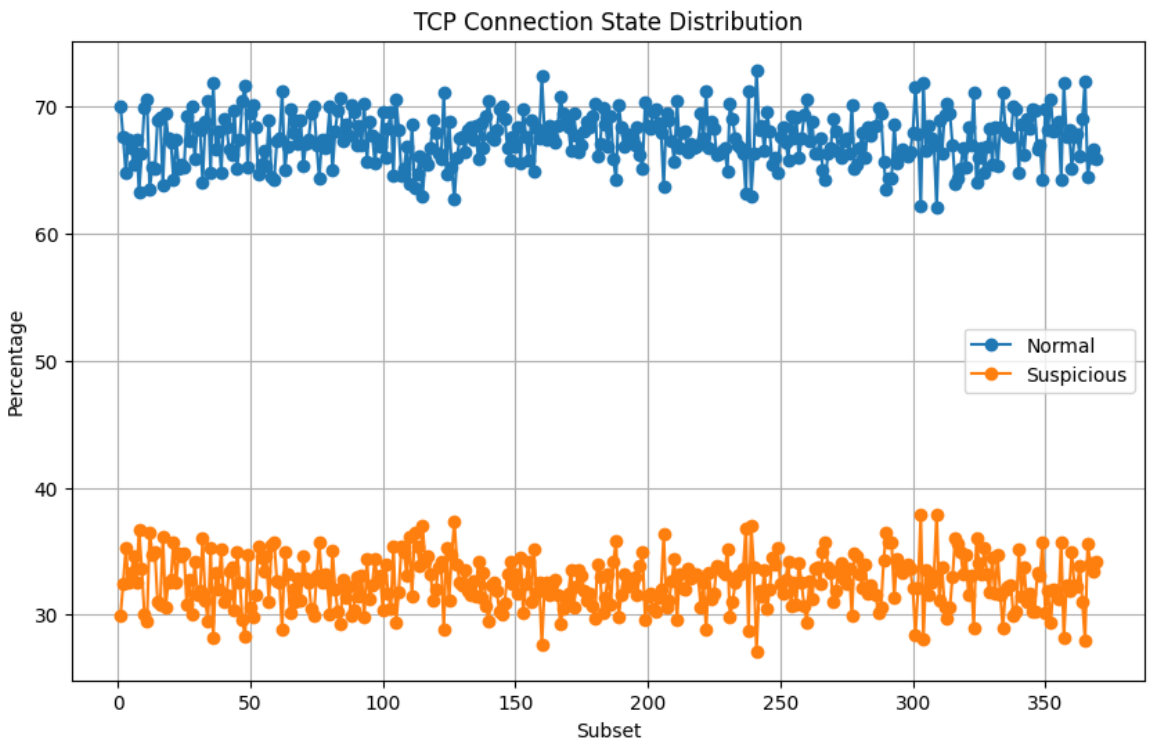

Figure 3: Percentage of TCP conncection state

4.1.2 UDP Flooding Analysis

For UDP traffic, volumetric statistics show more pronounced differences between DoS/DDoS and "Other" traffic.

The mean number of **destination packets** for DoS/DDoS traffic is approximately 1.1 packets per subset, compared to approximately 0.5-0.6 for "Other" traffic.

The computed destination packet threshold is 0.86, indicating that a substantial portion of DoS/DDoS traffic exceeds the typical packet volume observed in non-flooding traffic.

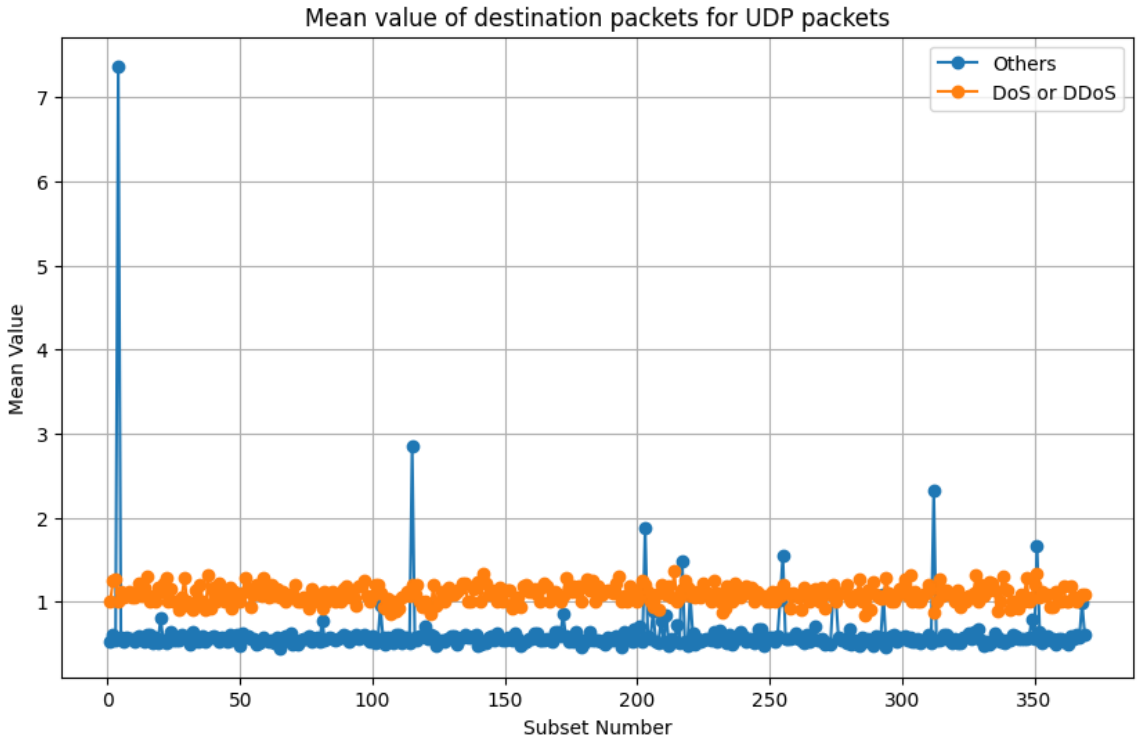

Figure 4: Mean value of destination packets for UDP packets

The mean number of **source packets** for DoS/DDoS traffic remains close to 1.0–1.2, while Other traffic generally exhibits higher variability.

A source packet threshold of 1.48 further supports the separation between flooding and non-flooding behavior.

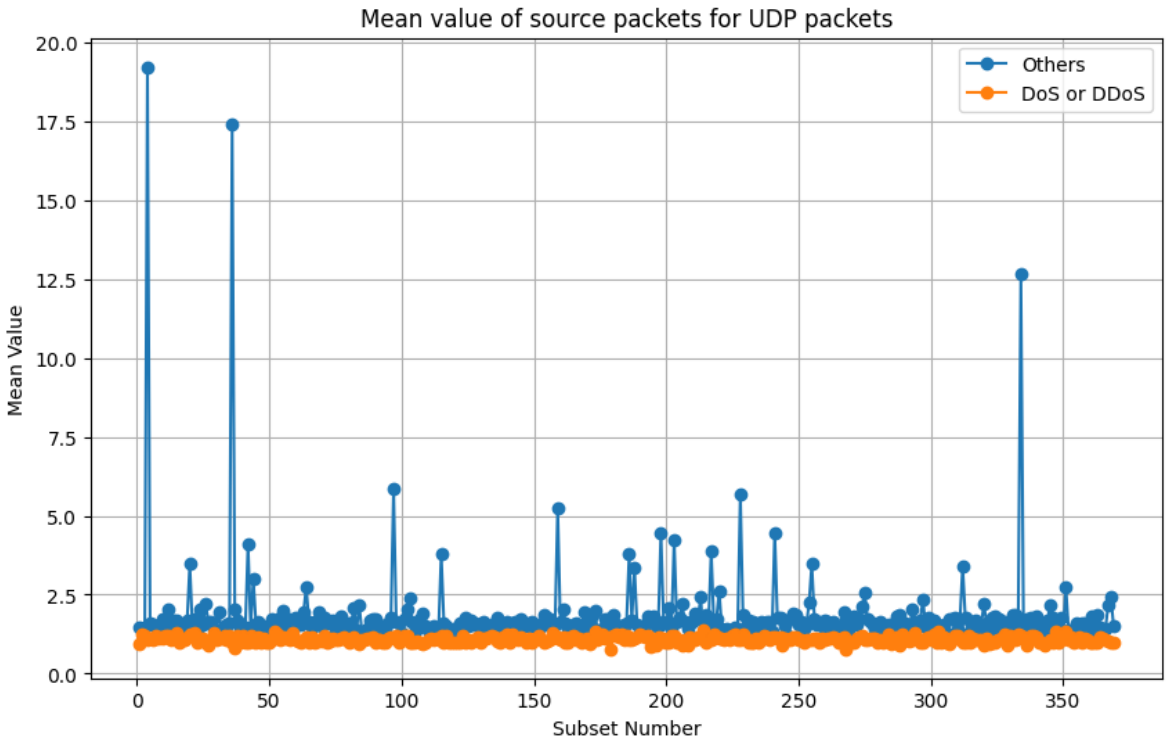

Figure 5: Mean value of source packets for UDP packets

Regarding byte-level statistics, the mean number of **destination bytes** for DoS/DDoS traffic is typically around 60–80 bytes, whereas "Other" traffic frequently reaches 180–200 bytes, with occasional spikes above 1000 bytes. The computed destination byte threshold is 124.39 bytes, which lies between these two distributions.

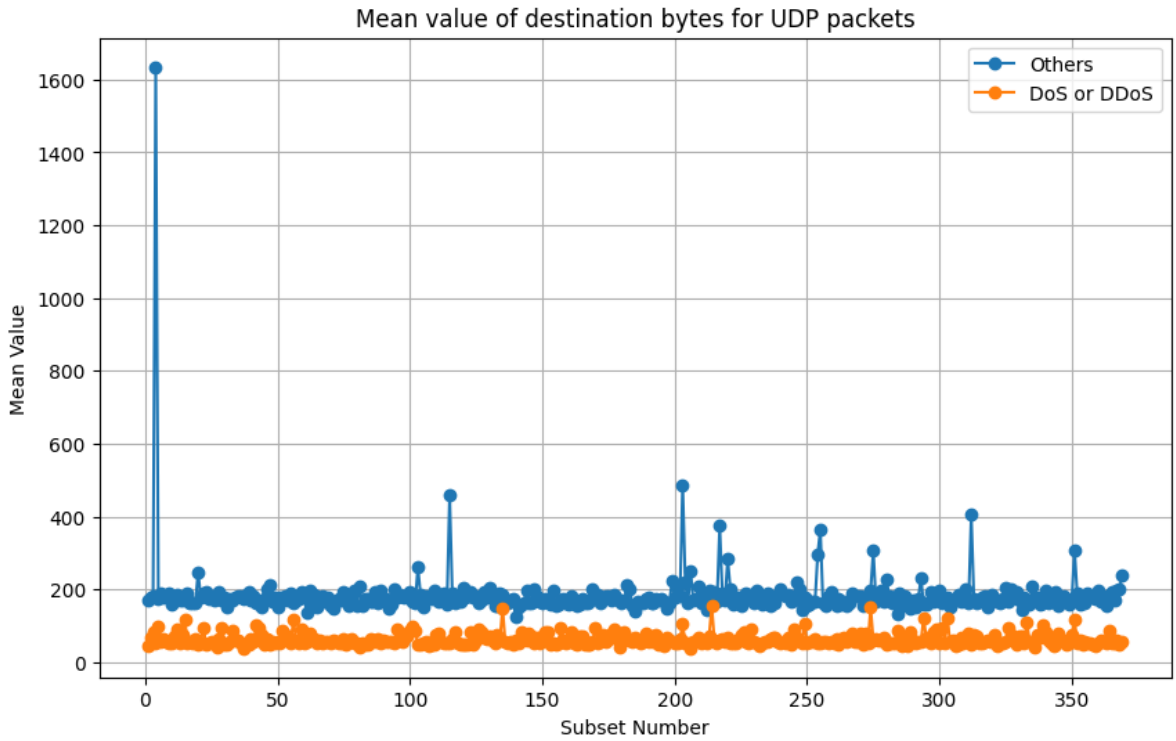

Figure 6: Mean value of destination bytes for UDP packets

Similarly, the mean number of **source bytes** for DoS/DDoS traffic remains relatively stable at low-to-moderate values, while "Other" traffic displays strong variability and extreme outliers.

The source byte threshold of 139.81 bytes further captures this separation.

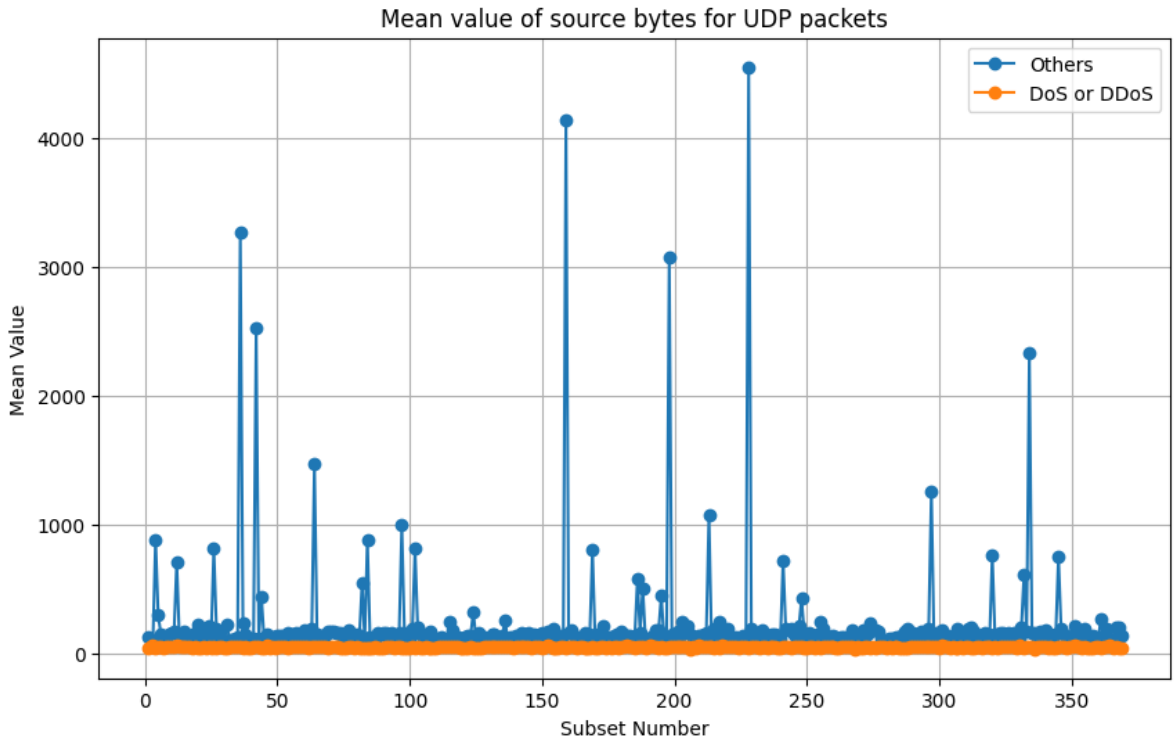

Figure 7: Mean value of source bytes for UDP packets

Furthermore, threshold-based percentage analysis shows that approximately 47% of UDP packets exceeds the destination packet threshold, while 78.7%, 67.6%, and 91.5% fall below the thresholds for source packets, destination bytes, and source bytes, respectively.

These distributions confirm that UDP flooding attacks generate a high volume of small packets, producing distinctive volumetric signatures that can be effectively exploited for detection.

4.2 Attack-Type Comparative Analysis

When comparing volumetric statistics across multiple attack types (DoS/DDoS, Scanning, Brute Force, WebAttack, and Ransomware), DoS/DDoS traffic shows the highest mean values and the largest variability.

Looking, for example, at the "Mean Source Packets per Attack Type" plot we can see that:

- DoS/DDoS traffic exhibits mean source packet values reaching peaks above 100 packets per subset in extreme cases.
- Brute Force and WebAttack traffic generally remain below 10–20 packets per subset.
- Ransomware and Scanning traffic display lower and more stable mean packet and byte counts.

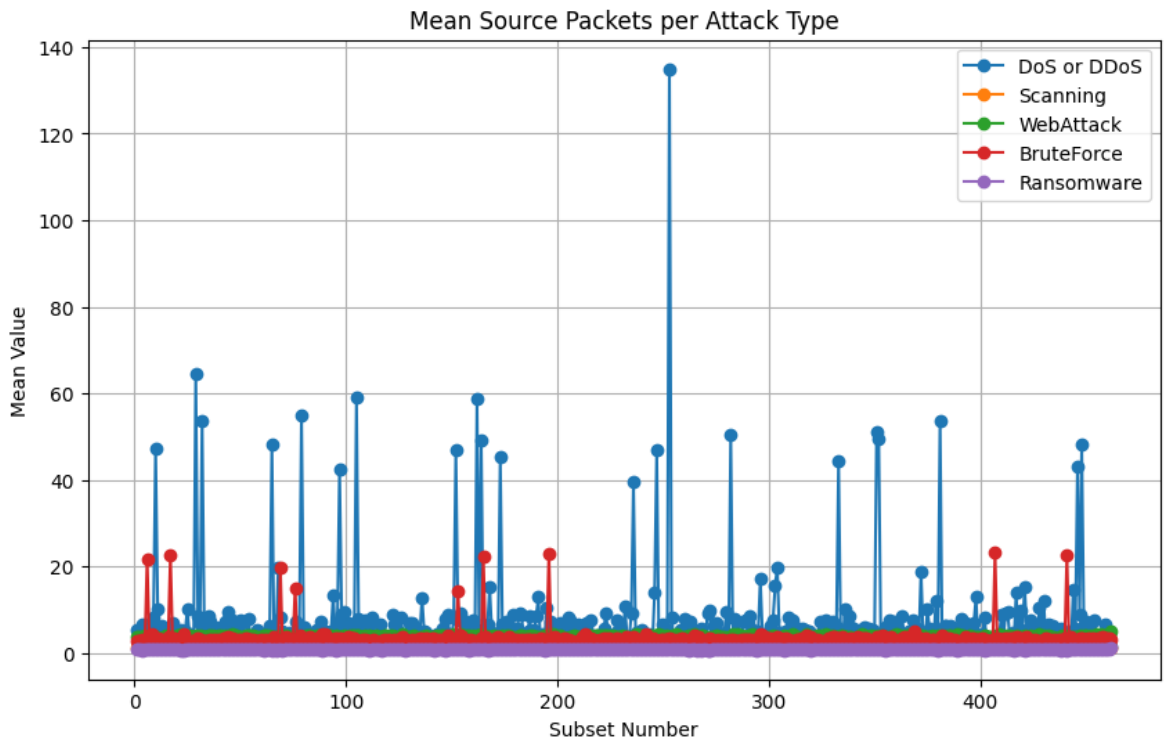

Figure 8: Mean source packets per attack type

These differences confirm that flooding-based attacks are primarily volumetric in nature, while other attack categories rely on targeted interactions rather than sustained high-rate traffic.

Similar patterns can also be observed when analyzing the other volumetric features, namely destination packets, destination bytes, and source bytes.

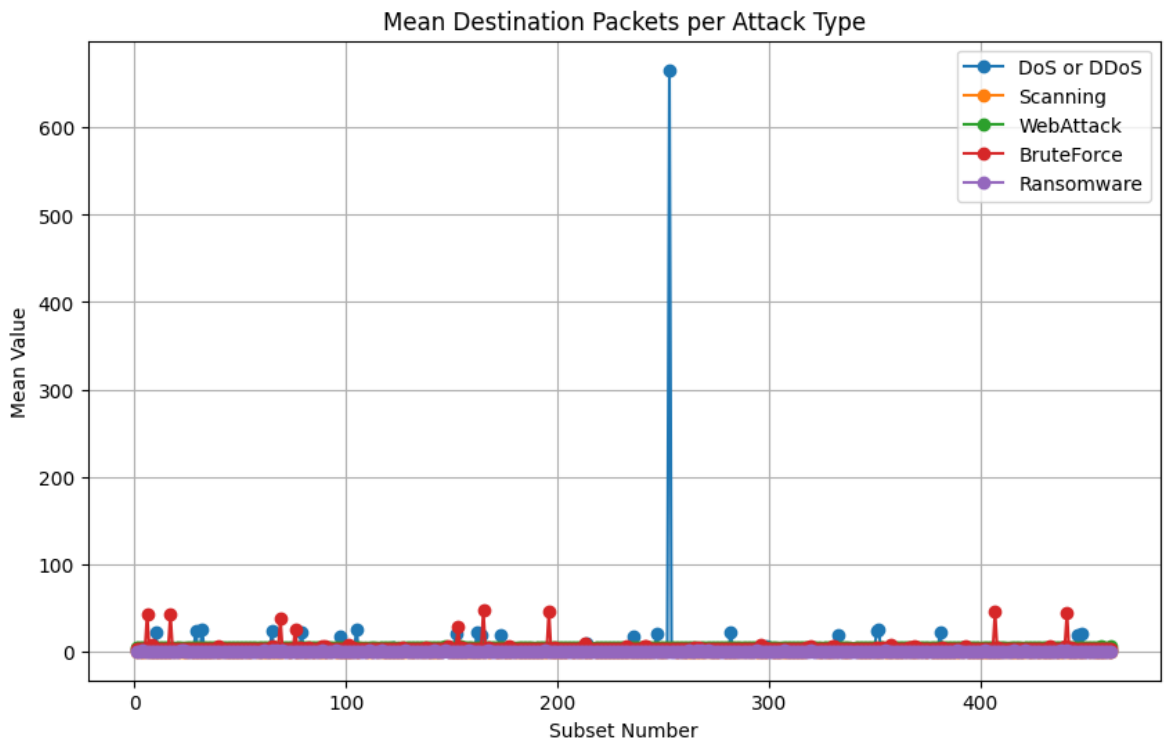

Figure 9: Mean destination packets per attack type

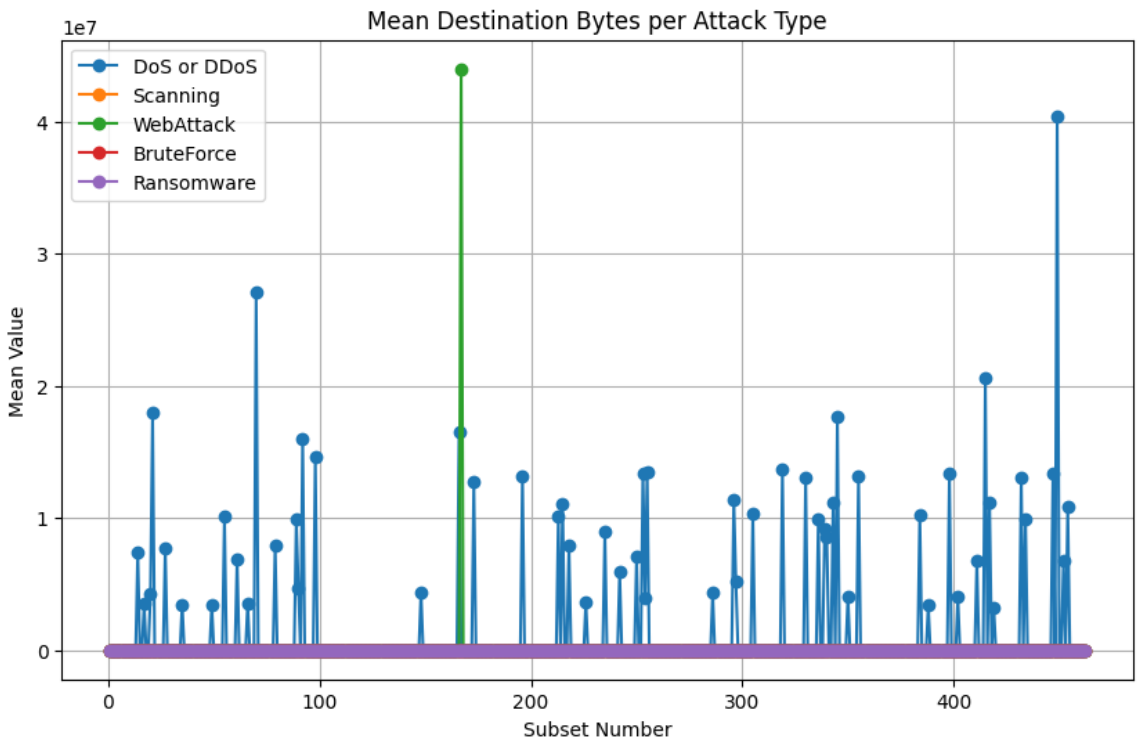

Figure 10: Mean destination bytes per attack type

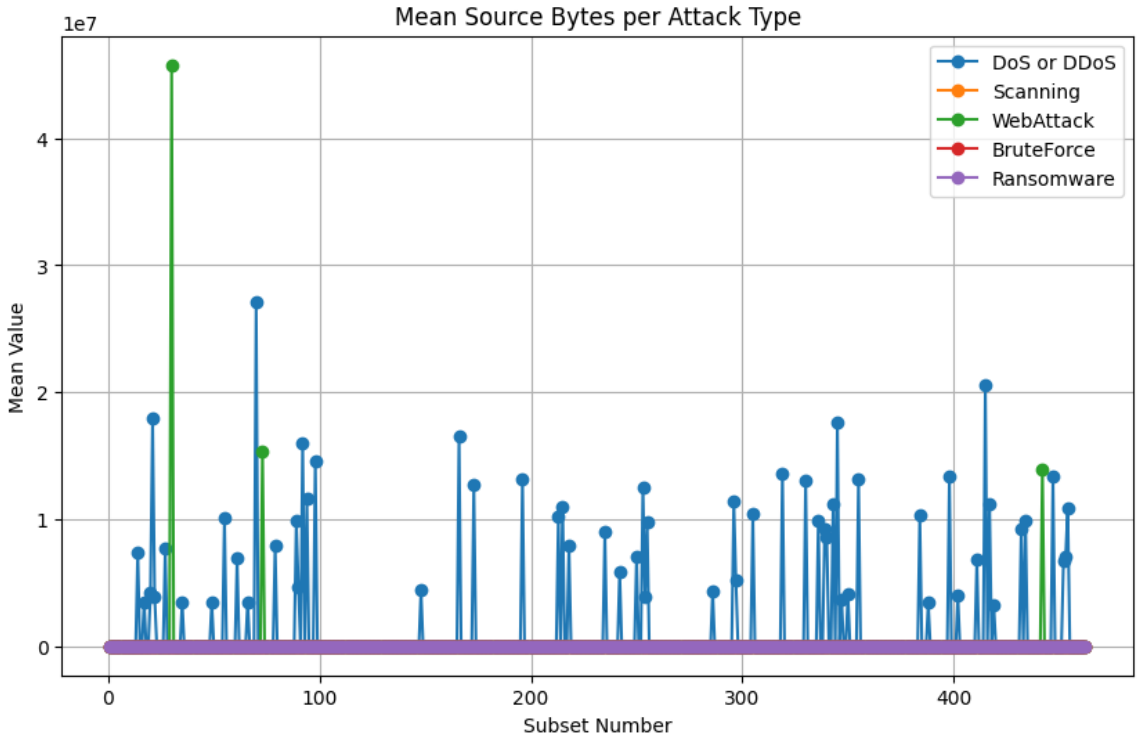

Figure 11: Mean source bytes per attack type

The corresponding plots show that DoS/DDoS traffic consistently produces the highest peaks and the largest variability across subsets, while the other attack types generally remain at lower and more stable values, with only occasional moderate spikes. These results further confirm the volumetric nature of flooding attacks, in contrast with other attack categories that generate more limited and controlled traffic patterns.

4.3 Machine Learning Classification Results

Three supervised machine learning models were evaluated: Logistic Regression, Support Vector Machine (SVM), and Random Forest. Performance was measured using accuracy, recall, and F1-score.

4.3.1 Results on Imbalanced Dataset

Before applying any balancing technique, the three classifiers were first trained and evaluated on the original dataset, preserving its natural class distribution. This baseline evaluation aims to analyze how each model behaves in the presence of strong class imbalance and to highlight the limitations that arise when minority-class samples (DoS/DDoS traffic) are underrepresented.
Below is the results table of the analysis:

Model	**Accuracy**	**Recall**	**F1-score**
Random Forest	0.9582	0.5547	0.6935
SVM	0.9148	0.0022	0.0044
Logistic Regression	0.9148	0.0021	0.0041

Table 2: Performance comparison of machine learning models with imbalanced dataset

As we can see, Random Forest achieves the highest overall accuracy (95.82%) and significantly outperforms the other models in terms of recall and F1-score. This indicates a substantially better capability to correctly identify DoS/DDoS traffic.

Although Logistic Regression and SVM reach high accuracy values, their extremely low recall and F1-score demonstrate that they almost always classify samples as belonging to the majority class "Other".

This behavior is a direct consequence of the imbalanced nature of the dataset, where benign and non-DoS traffic largely outnumbers DoS/DDoS samples.

Random Forest, being an ensemble of decision trees, is able to capture complex and non-linear relationships among features and is more robust to class imbalance; as a result, it provides a more reliable detection of flooding attacks.

These findings confirm that ensemble-based models are better suited for intrusion detection tasks involving heterogeneous traffic and unbalanced datasets.

4.3.2 Results after SMOTE balancing dataset

To mitigate class imbalance, the **Synthetic Minority Over-sampling Technique** (SMOTE) was applied to the training set.

SMOTE [4] is an oversampling technique that generates synthetic samples for the minority class rather than simply duplicating existing ones.

This approach reduces overfitting and improves the representativeness of the minority class in the feature space.

Mathematically, given a minority class sample x_i and one of its $k-$nearest neighbors x_j, a new synthetic sample x_{new} is generated according to the equation:

$$x_{\text{new}} = x_i + \lambda \cdot (x_j - x_i)$$

where $\lambda \in [0, 1]$ is a randomly generated number.

After applying SMOTE, the dataset becomes perfectly balanced between the two classes (DoS/DDoS and Other) with 269k samples each, allowing the models to learn decision boundaries that are not biased toward the majority class.

The test set was intentionally kept unchanged in order to preserve a realistic and unbiased evaluation scenario.
The performance obtained after SMOTE balancing is summarized below:

Model	**Accuracy**	**Recall**	**F1-score**
Random Forest	0.9192	0.9795	0.6740
SVM	0.1228	0.2623	0.0486
Logistic Regression	0.3856	0.7550	0.1734

Table 3: Performance comparison of machine learning models after SMOTE balancing

After balancing, all models exhibit a substantial improvement in recall compared to the imbalanced scenario, confirming that class imbalance was a major factor limiting the detection of DoS/DDoS traffic.

Random Forest maintains a very high precision (91.92%) while achieving an excellent recall (97.95%), indicating that the vast majority of flooding attacks are correctly identified with a low false-positive rate.

The resulting F1-score (0.6740) confirms a strong overall balance between accuracy and recall.

Logistic Regression benefits noticeably from SMOTE, reaching a recall of 75.50%, which represents a significant improvement over the imbalanced case. However, its moderate accuracy (38.56%) suggests a higher number of false positives.

SVM shows only limited improvement and continues to underperform, indicating difficulty in modeling the complex and non-linear patterns associated with flooding behavior.

Overall, these results demonstrate that data balancing is essential for intrusion detection tasks and that ensemble-based models such as Random Forest are particularly well-suited for handling heterogeneous network traffic and complex attack patterns.

5 Conclusions and future work

This project addressed the problem of detecting Denial of Service (DoS) and Distributed Denial of Service (DDoS) attacks in network traffic, with particular emphasis on IoT-oriented environments, by combining statistical traffic analysis with supervised machine learning techniques.

Using the TON_IoT_Train_Test_Network dataset, the study analyzed both behavioral and volumetric characteristics of network flows in order to identify discriminative patterns associated with flooding-based attacks.

The statistical analysis revealed clear and consistent differences between DoS/DDoS traffic and other types of network activity.

For TCP traffic, abnormal connection states were shown to be strong indicators of flooding behavior, as reflected by stable proportions of suspicious connections across traffic subsets.

For UDP traffic, where connection state information is not available, volumetric features such as packet counts and byte distributions proved effective in characterizing flooding activity.

Furthermore, a comparative analysis across multiple attack types (Scanning, Brute Force, WebAttack, and Ransomware) demonstrated that flooding-based attacks are predominantly volumetric in nature, whereas other attacks typically involve more targeted and lightweight communication patterns.

The machine learning evaluation confirmed that model selection plays a crucial role in intrusion detection performance. Although Logistic Regression and Support Vector Machine achieved relatively high accuracy values on the original imbalanced dataset, their very low recall and F1-score indicate a strong bias toward the majority class, leading to poor detection of DoS/DDoS traffic.

After applying the Synthetic Minority Over-sampling Technique (SMOTE) to balance the training data, all models exhibited a substantial improvement in recall, confirming that class imbalance was a major limiting factor. Nevertheless, Random Forest consistently achieved the best overall performance, providing both high precision and significantly improved recall and F1-score.

These results highlight the effectiveness of ensemble-based approaches in capturing non-linear relationships, as well as the importance of data balancing techniques when dealing with highly skewed security datasets.

Overall, the results suggest that combining statistical traffic analysis with ensemble machine learning models represents an effective strategy for DoS/DDoS detection. Statistical analysis offers interpretability and insight into traffic behavior, while machine learning enables scalable and automated classification in complex environments.

Several limitations remain. The experiments were conducted using a single dataset and under offline conditions. Moreover, the focus of this work was

primarily on binary classification between DoS/DDoS and other traffic.

Future work may explore multi-class classification, additional datasets, advanced imbalance-handling techniques, and deep learning-based approaches. Real-time deployment and evaluation in operational environments also represent important directions for further research.

In conclusion, this work demonstrates that meaningful and reliable DoS/DDoS detection can be achieved through careful feature selection, statistical characterization of traffic, and appropriate machine learning modeling, providing a solid foundation for future research and practical deployment in modern network and IoT security systems.

6 How to run the code

This project was developed and tested using Google Colab. The following steps describe how to correctly run the code and reproduce the experiments.

1. Download the dataset from Kaggle at the following link: `https://www.kaggle.com/datasets/alaaelmor/ton-iot-train-test-network`
2. Open Google Colab
3. Upload the dataset to Colab
4. Run the notebook sequentially

6.1 Note on reproducibility

Due to the stochastic nature of certain operations (such as dataset shuffling, train-test splitting, and SMOTE sample generation), small variations in the reported performance metrics may occur across different executions.

Although a fixed *random_state* was used to improve reproducibility, minor fluctuations in precision, recall, and F1-score are expected and do not affect the overall conclusions of the study.

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